

EFS-02- Statistical Arbitrage and Pair trading

Overview

This session will take you through basics of the Statistical Arbitrage Strategy along with implementing a pairs trading strategy on EWA and EWC in Python. In this lecture, faculty with his practical knowledge and hands-on experience will answer all your doubts/queries/concerns related to Statistical Arbitrage.

This is an interactive lecture; hence, feel free to ask your questions, doubts, etc. to the faculty during the lecture. You may also write back to us via LMS in case you need any further assistance on any of the topics covered in the lecture.

Key Takeaways:

- Pair Trading Methodology to be discussed in detail in terms of identification of probable pairs, entry-exit and back-testing
- In-depth understanding of co-integration methodology
- Discussion on position sizing and risk management

Pre-requisites:

- Elementary statistics such as z-score, mean, and standard deviation
- Linear regression
- Bollinger bands
- Hypothesis testing
- Python: pandas
- Read articles on pair trading from the internet to get a general idea of the strategy

Environment setup requirements:

- Jupyter notebook
- Python Package
1. Statsmodels

Downloadable Files:

- Please download the file set for the class 'EFS-02-Files' available in course "Equity, FX, & Futures Strategies".

Practical use of the topics learned in this session:

This session is the building base of your Statistical Arbitrage Strategy knowledge which is very important in conducting any research or planning for the strategy.

Post this lecture, you will be in a position to understand:

- Pair Trading Methodology – Identification of probable pairs, entry-exit and back-testing
- In-depth understanding of Stationarity & cointegration
- Position sizing and risk management

Recommended time for the session & related coursework - 10 hours

Additional Read

Blog:

[Pair Trading – Statistical Arbitrage On Cash Stocks](#)

[EPAT Final Project by Jacques Joubert – Statistical Arbitrage Strategy in R](#)

[Statistical Arbitrage: Pair Trading In The Mexican Stock Market](#)